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backend=biber, style=numeric, sorting=none]biblatex Safeguarding a Flagship Species: Integrated Surveillance of Cross-Species Pathogen Transmission in Giant Panda Ecosystems

Xiaoli Tang[1], Xiaoli Sun[1], Yi Peng[1], Xiaoye Hao[1], Rong Dong[2], le wang[3], Chengdong Wang[4], Xiangdong Wu[1], Zheng Chen [1], and Wenbo Zhang[1]

[1]Jiangxi Agricultural University
[2]Research Center for Qinling Giant Panda, Shaanxi Academy of Forestry
[3]China West Normal University
[4] China Conservation and Research Centre for the Giant Panda， Key Laboratory of SFGA on the Giant Panda, Chengdu, China

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Abstract

Emerging infectious diseases, driven by increasing interactions among humans, wildlife, and livestock pose an escalating threat to global health, biodiversity, and economies. As a flagship endangered species, the giant panda plays a pivotal role in biodiversity conservation in China. This review synthesizes current knowledge on pathogens threatening giant panda health, including viruses, bacteria, and parasites alongside their potential transmission pathways within nature reserves. We emphasize the roles of domesticated animals, sympatric wildlife, and ectoparasites as reservoir hosts or vectors. Special focus is placed on cross-species transmission dynamics and the critical need for integrated monitoring systems utilizing metagenomics and viromics. We propose a framework for establishing early warning systems and surveillance networks at the domestic-wild animal interface to enhance pathogen detection, disease prevention, and biodiversity conservation.

Short title:**Integrated Pathogen Surveillance in Giant Panda Ecosystems**

Xiaoli Sun[1#], Yi Peng[1#], Xiaoye Hao[1#], Rong Dong[2], Le Wang[3], Chengdong Wang[4], Xiangdong Wu[1], Zheng Chen[1], Wenbo Zhang[1] and Xiaoli Tang[1*]

[1] College of Animal Science and Technology, Jiangxi Agricultural University, Nanchang 330045, China

[2] Jiangxi Provincial Key Laboratory of Conservation Biology, College of Forestry, Jiangxi Agricultural University, Nanchang 330045, China

[3] Key Laboratory of Southwest China Wildlife Resources Conservation (Ministry of Education), China West Normal University, Nanchong, 637001, China

[4] Chengdu Research Base of Giant Panda Breeding, Chengdu 610081, China.

[5] Research Center for Qinling Giant Panda, Shaanxi Academy of Forestry, Xi'an 710000, China

***** Correspondence: m15600912196@163.com

[#] These authors contributed equally to this work.

Impact Statement:

Integrated pathogen surveillance in panda reserves is vital to prevent disease spillover and protect endangered species and ecosystem health.

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Abstract

Emerging infectious diseases, driven by increasing interactions among humans, wildlife, and livestock pose an escalating threat to global health, biodiversity, and economies. As a flagship endangered species, the giant panda plays a pivotal role in biodiversity conservation in China. This review synthesizes current knowledge on pathogens threatening giant panda health, including viruses, bacteria, and parasites alongside their potential transmission pathways within nature reserves. We emphasize the roles of domesticated animals, sympatric wildlife, and ectoparasites as reservoir hosts or vectors. Special focus is placed on cross-species transmission dynamics and the critical need for integrated monitoring systems utilizing metagenomics and viromics. We propose a framework for establishing early warning systems and surveillance networks at the domestic-wild animal interface to enhance pathogen detection, disease prevention, and biodiversity conservation.

Keywords

Giant panda; Domestic-wild animal interface; Disease cross-species transmission; Pathogen surveillance; Metagenomics

1. Introduction

In recent decades, emerging infectious diseases, such as plague, Middle East Respiratory Syndrome, and COVID-19 have presented unprecedented challenges to wildlife conservation, public health, biodiversity, and socio-economic development. These threats stem from complex factors including pathogen evolution, climate change, land-use alterations, economic development, and human population expansion(1). The degradation and fragmentation of natural habitats have intensified interactions between humans, domestic animals, wildlife, and vectors, creating ideal conditions for cross-species transmission of infectious agents. Research indicates that wild mammals harbor at least 878 viral species, including 170 with zoonotic potential, and livestock carry 288 viruses, of which 117 are zoonotic(2). Notably, pathogen transmission at the domestic–wild animal interface is frequently bidirectional(3). It is estimated that over 70% of zoonotic diseases originate from wildlife (4). Furthermore, approximately 1.7 million undiscovered viral species exist in wildlife globally, with 540,000 to 850,000 potentially capable of infecting humans directly or via domestic animals (5). Conversely, infectious diseases have also emerged as a primary threat to wildlife survival. Human activities, including animal farming and global trade, exacerbate pathogens spread. Notably, a large proportion of livestock pathogens (77%), and an even higher proportion of carnivore pathogens (91%), can infect multiple hosts, including wildlife(3). For instance, during 2016-2017, the Peste-des-petits-ruminants' virus spilled over from livestock to the critically endangered Saiga antelope (Saiga tatarica), causing 80% mortality(6). Since 2007, African Swine Fever Virus (ASFV) has spread across Europe and Asia through pig and pork trade, subsequently spilling over to wild boar populations and threatening Southeast Asia's

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11 endemic wild pig species(7). Eradicating livestock-derived pathogens once established in wildlife populations remains extremely challenging. Therefore, preventing pathogen spillover at the domestic and wildlife interface is essential for wildlife conservation, public health, and economic stability (3). However, current understanding of diseases dynamics and transmission mechanisms at this interface remains limited. The absence of effective early warning systems and diagnostic tools hinders the control of emerging infectious diseases. This underscores the urgent need for enhanced surveillance, risk assessment, and control strategies for these diseases in ecosystems shared by domestic and wild species. The giant panda (*Ailuropoda melanoleuca*), an endemic and endangered species in China, is a global flagship species for biodiversity conservation(8). Due to successful conservation efforts, the IUCN reclassified its status from "endangered" to "vulnerable" in 2016(9). According to the Fourth National Survey on Giant Pandas, an estimated 1,864 wild individuals are confined to six mountain ranges: the Qinling, Minshan, Qionglai, Daxiangling, Xiaoxiangling, and Liangshan Mountains(10).In 2021, the Giant Panda National Park was officially established across primary panda habitats in Sichuan, Shaanxi, and Gansu provinces, covering 22,000 square kilometers. The park established ecological corridors connecting 13 local panda populations, thereby protecting 71.9% (1,340 out of 1,864) of the wild giant pandas nationwide. This park harbors numerous endemic, rare, and endangered animals and is critical for Asian biodiversity conservation. Significant human populations reside within and around these reserves, with villages often overlapping reserve boundaries, some situated near core panda habitats (11). This proximity facilitates direct and indirect contact among residents, domestic animals (livestock, dogs, cats), giant pandas, and sympatric wildlife, substantially increasing pathogen transmission risks. Documented cases include giant panda deaths from Canine Distemper Virus (CDV) transmitted by domestic dogs(12), the first reported cases of *Hepatozoon* sp. infection and human-origin pH1N1 influenza infection in pandas(13) , and the potential transmission of *Enterocytozoon bieneusi* from pandas to humans(14). These incidents highlight the critical need to establish baseline data on pathogens carried by giant pandas, sympatric wildlife, domestic animals, and vectors within panda reserves. Research into transmission mechanisms and the creation of a wildlife pathogen community database are essential for supporting disease prevention and control efforts for giant pandas and other endangered wildlife, ultimately contributing to species conservation, biodiversity maintenance, and human health. This review first comprehensively summarizes currently reported pathogens (viruses, bacteria, parasites) threatening giant panda health, detailing their prevalence, spatiotemporal distribution, morbidity, and mortality. Second, it synthesizes current knowledge on transmission pathways involving these pathogens, elucidating the impact of pathogens carried by sympatric animals, domestic animals, and vectors on giant pandas.

2. Pathogens Threatening the Health of Giant Pandas

Given the giant panda's vulnerable conservation status, the presence of any pathogen warrants consideration for its clinical and conservation management implications. Although some pathogens may be asymptomatic in healthy captive individuals, they can cause severe consequences in juveniles, elderly animals, or immunocompromised individuals, disproportionately affecting small and isolated populations(15, 16). Moreover, co-infections may potentiate the pathogenic effects of individual microorganisms(17). Currently, the threat of infectious diseases within the giant panda's natural habitat is substantial. Panda reserves are often intricately linked with human settlements. For example, Wolong Nature Reserve, protects approximately 2,000 km^2 but is surrounded by 1,436 households(11) . Liziping National Nature Reserve, a key site for panda translocation and reintroduction, is surrounded by villages housing over 7,000 residents(18). Road construction, human settlements near panda habitat, and the presence of domestic pets and livestock directly increase the livestock-wildlife interface, promoting disease transmission. Furthermore, habitat loss and fragmentation concentrate pandas within remnant bamboo forests(19). Concurrently, panda density has increased fourfold as populations have grown(20), elevating pathogen prevalence and infection intensity within panda populations and presenting unprecedented challenges for disease control. While the establishment of panda reserves protects this flagship species and benefits sympatric mammals through umbrella and ecosystem-wide conservation effects(21), the broad host range of most pathogens means these sympatric wildlife species themselves harbor and transmit pathogens, posing a potential threat for disease outbreaks. Additionally, the current wild giant panda population is fragmented into 33 isolated subpopulations. Small, isolated populations face

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increased risks of inbreeding depression, potentially leading to reduced disease resistance and insufficient immunity against pathogens(22) (23). Alarmingly, 15 subpopulations face a >90% extinction risk. Translocation conservation and reintroduction programs are crucial for panda population maintenance but also directly facilitate disease transmission between regions. Consequently, infectious diseases, including viral, bacterial, and parasitic infections can rapidly spread among wildlife, domestic animals, and even humans within panda reserves, causing large-scale outbreaks and mortality, posing significant threats to human health and biodiversity.

2.1 Viral Diseases

Although clinical data on viral infections in giant pandas remain limited, several viruses have been associated with disease and mortality in both captive and wild individuals. These include canine distemper virus (CDV), canine parvovirus (CPV), feline panleukopenia virus (FPV), adenovirus, coronavirus, parainfluenza virus, rabies virus, circovirus, and influenza viruses(15, 24). Among these, CDV, CPV, FPV, and Canine Coronavirus have been directly linked to morbidity and mortality in giant pandas(25, 26) (Table 1).CDV poses the most severe threat to captive pandas. During an outbreak in 2014–2015, five of six infected pandas died at the Shaanxi Rare Wildlife Rescue and Breeding Center(27). Giant panda habitats in Shaanxi, Sichuan, and Gansu provinces border three CDV high-prevalence areas; any CDV outbreak in these regions could endanger panda populations(26). Rotavirus causes refractory diarrhea and chronic malnutrition syndrome in weaned cubs, parvovirus causes viral enteritis, and Influenza A virus also impacts panda health. The multi-host nature of these pathogens threatens not only pandas but also other wildlife and humans. In 2023, a panda-derived FPV strain causing severe or fatal symptoms was isolated; this strain could infect cell lines from other mammalian species, including humans(25). Beyond recognized viruses, numerous undiscovered viruses likely exist. Viral metagenomics has recently facilitated the discovery of numerous animal viruses, providing insights into virome composition and aiding in identifying etiological agents, detecting zoonoses, and identifying novel and emerging viruses. Using viromics techniques, novel picornaviruses, anelloviruses (19 distinct variants), and papillomaviruses have been detected in giant pandas(28). While representatives of these virus families were also detected in healthy pandas, a higher proportion was found in diseased individuals, suggesting a potential association with illness. Viral presence does not necessarily indicate disease but may represent a degree of symbiotic interaction between virus and host(29). However, the stability of this virome-host association is uncertain; waning host immunity may increase viral replication (30) , leading to higher viral loads in immunocompromised pandas and further deteriorating their condition[25]. Nevertheless, the transmission pathways, host adaptation variations, pathogenicity, and distribution patterns among sympatric animals of these viruses require further investigation.

2.2 Bacterial Infections and Antimicrobial Resistance

Gastrointestinal bacterial infections are considered a leading cause of mortality in giant pandas, as alterations in gut microbial community can lead to digestive disorders and the development of underlying diseases(31). Common bacterial pathogens include *Escherichia coli* , *Klebsiella pneumoniae* , *Campylobacter jejuni* ,*Arizona spp.* , *Pseudomonas aeruginosa* , *Clostridium spp.* , and *Proteus mirabilis(32)* . The widespread use of antibiotics has led to an increased prevalence of antimicrobial resistance (AMR). Antibiotic resistance genes (ARGs) can be transmitted via mobile genetic elements (MGEs) such as plasmids and integrons, facilitating horizontal gene transfer among bacteria. Genes like*tolC* , *mepA* , and *mdtA* have been implicated in multi-drug resistance in giant panda-associated isolates, leading to the emergence of multidrug-resistant bacteria and posing significant risks to panda health (32). In 2015, a giant panda at Shanghai Wildlife Park died from infection with multidrug-resistant bacteria, such as *E. coli* , *P. aeruginosa* , and *Enterococcus faecalis* . The *E. coli* strain SH-YH-DH carried a self-transmissible plasmid encoding multiple resistance genes (*bla* CTX-M-55, *bla* TEM-1, *sul1* , *floR* , *strB* , *aac(6')Ib* , *tetA/R*) and exhibited resistance to gentamicin, cefotaxime, ampicillin, chloramphenicol, trimethoprim-sulfamethoxazole, and tetracycline (33). The giant panda gut microbial community is a significant reservoir of ARGs, including high-risk resistance genes(34). Analysis of fresh fecal samples from captive pandas at the Chengdu Research Base of Giant Panda Breeding using HT-qPCR, detected 47 ARGs and 8 MGEs among isolated*K. pneumoniae* strains; some

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strains carried multiple ARGs and MGEs, highlighting the importance of monitoring and researching AMR in wildlife populations(32). Furthermore, virulence genes (e.g., *magA* , *uge* , *ycf* , *entB* , *kpn* , *allS* , *wabG* , *mrk* , *iro*) and heavy metal tolerance genes are enriched in the panda microbiota, contributing to various symptoms and aiding bacterial immune evasion through different mechanisms(32). Compared to wild pandas, captive populations exhibit significantly higher relative abundances of high-risk ARGs, virulence genes, and heavy metal tolerance genes(34, 35). If captive individuals are reintroduced into the wild, this could have catastrophic consequences for wild panda populations, other wildlife, and ecosystems (35). Besides the impact of captive-origin pathogens on wild populations, pathogens from wild populations also threaten captive pandas. In the 1980s, some captive pandas died from hemorrhagic enteritis caused by enteroinvasive *E. coli* O152 introduced by a rescued wild panda. Moreover, ARGs found in giant pandas and humans show close genetic similarity (36), emphasizing the significant impact of human activities on ARG prevalence in natural environments and the potential health risks ARGs pose to pandas. The environment is a key factor influencing AMR in the panda gut microbiome, as multiple ARGs and MGEs are shared between the gut and the habitat environment. In-depth research on the distribution of gut microbiota and resistance genes among sympatric animals of the giant panda is crucial for species conservation efforts(34)(Table 2).

2.3 Parasitic Infections

Parasitic diseases have long been recognized as major threats to the survival of wild giant pandas (20). A retrospective analysis from 1971 to 2005 showed that parasitic infections accounted for over 50% of deaths in free-ranging pandas. To date, at least 35 parasite species have been documented in pandas, belonging to nematodes (6 species), trematode (1 species), cestodes (2 species), protozoans (9 species), and ectoparasites (13 ticks, 2 mites, 2 fleas) (37). Clinical significant cases commonly involve *Baylisascaris schroederi* (*B. schroederi,* ascarid roundworm), *Chorioptes panda* (mange mite), and Ixodid ticks(23) . *B. schroederi* has the highest prevalence (7%–88%) and mortality rate (50%), and is considered the primary cause of parasitic mortality(20, 23). Its larvae can migrate through visceral tissues, while adults can cause intestinal obstruction, pancreatitis, and hemorrhagic enteritis. One individual was found to harbor over 3,200 adult worms(38). Other prevalent endoparasites include *Ogmocotyle sikae* (trematode, prevalence 0.5%-100%), *Ancylostoma ailuropodae* (hookworm, prevalence 93.3%), *Enterocytozoon bieneusi* (microsporidian, zoonotic, prevalence 8.7-34.5%), *Toxoplasma gondii* (protozoan, zoonotic, prevalence 100%), and *Hepatozoon* sp. (protozoan, prevalence 73.9%). *A. ailuropodae* inhabits the small intestine, causing intestinal mucosal hemorrhage and inflammation(39). *O. sikae* also resides in the small intestine, leading to multiple mucosal hemorrhagic spots and digestive dysfunction (37). *Cryptosporidium* spp. and *Enterocytozoon bieneusi* infect the intestines, primarily causing intestinal tissue damage, diarrhea, and weight loss (40, 41).

Besides endoparasites, ectoparasite infections are frequently reported and represent the second most prevalent parasitic problem(37). Ticks (prevalence 2%-100%), *Haemaphysalis ailuropodae* (prevalence 50%), and *Chorioptes panda* (prevalence 66.7%-100%) cause significant harm. Infections with *Demodex ailuropodae* and *C. panda* cause sarcoptic mange, with prevalence ranging from 66.7% (6/9) to 100% (7/7)(37). Ticks are the second most important blood-feeding arthropods after mosquitoes. Thirteen tick species have been identified on wild giant pandas to date, belonging to the genera *Haemaphysalis* (9 species), *Ixodes* (3 species), and *Dermacentor* (1 species). Mixed tick infestations are common, with *Haemaphysalis flava* being the most prevalent in panda populations(23). These ectoparasites cause anemia, dermatitis, and secondary bacterial infections. Tick infestation is characterized by anemia, malnutrition, inflammation, and exhaustion in pandas(38). More importantly, tick-borne diseases can lead to devastating secondary infections by other pathogens (viruses, bacteria, parasites). Ectoparasites, particularly ticks, are proven vectors for vertebrate pathogens and facilitate interactions and genetic exchange among viruses(42). Using viral metagenomics, Rui Ma et al. identified 32 viruses in ticks collected from two giant pandas in the Daxiangling Nature Reserve; half showed homology to viruses carried by pandas and sympatric animals (e.g., red pandas), indicating frequent viral exchange between ticks and their hosts like pandas (42). Several phylogenetically classified viruses (Bunyaviruses, Hepacivirus-like, Circovirus-like) were relatively abundant, though whether these novel tick-associated viruses can replicate in ticks and be transmitted to host animals during blood-feeding

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requires further study. These findings expand our understanding of the role of panda-associated ticks in local ecosystems, particularly in virus acquisition and transmission, and lay the groundwork for assessing panda exposure risks to tick-borne viruses, reporting numerous cases of viral host-switching among sympatric species. Regarding bacteria, approximately 10% of all tick species (over 900) play significant roles in transmitting agents of diseases like borreliosis and rickettsiosis(42). However, detailed studies on tick-borne bacterial diseases within panda reserves are currently lacking. While a direct causal link between tick-borne pathogens and mammalian disease in pandas cannot be definitively established, it is evident that many viruses carried by ticks are genetically related to viruses found in pandas and other hosts sharing adjacent habitats(42). *Babesia* spp., tick-borne protozoan blood parasites, can cause hemolytic anemia, thrombocytopenia, lethargy, and splenomegaly in pandas(43). Ticks serve as vectors for *Babesia* , and babesiosis is primarily transmitted between hosts via tick bites in natural environments(44). Other parasites may act as opportunistic pathogens, causing disease or exacerbating conditions when immunity is compromised by viral or bacterial infections(43)(Table 3).

Regular deworming and environmental disinfection can effectively protect captive pandas from parasitic diseases. However, frequent drug overuse in captive settings has led to the development of drug resistance in parasites, particularly against common anti-*B. schroederi* anthelmintic. Furthermore, drug resistance genes carried by captive pandas can be disseminated through reintroduction programs (37). Tao Wang et al. found high levels of gene flow for *B. schroederi* among panda populations in different habitats, indicating the potential for rapid spread of anthelmintic resistance (23), posing a major obstacle to controlling this parasite and necessitating integrated management approaches. Developing sensitive, convenient and high throughput detection methods for panda parasites is another crucial aspect for assessing prevalence and distribution in captive and wild populations(23). Critically, the infectivity, pathogenicity, range, prevalence, and vectors for parasites in wild populations remain largely unknown. Further research is needed to elucidate pathogen-host-vector relationships and determine the necessity for targeted conservation management actions for this vulnerable species.

3. Pathogen Carriage by Domestic and Wild Animals in Giant Panda Nature Reserves

Numerous pathogens are transmitted to giant pandas through direct or indirect contact with free-roaming domestic animals (e.g., livestock, stray dogs and cats) and sympatric wildlife within nature reserves. These interactions create complex transmission networks and potential risks of cross-species disease outbreaks. Therefore, a comprehensive understanding of the spatial distribution of domestic animals, wildlife, and their associated pathogens is fundamental for assessing the risks and mechanisms of infectious disease transmission in giant panda nature reserves(Table 4).

3.1 Domestic Pets

Domestic dogs and cats, especially those free- roaming in or near giant panda reserves, represent significant sources of infectious agents. Their relatively high population densities, frequent contact with humans, and poor immunization increase the risk of introducing zoonotic and multi-host pathogens into wildlife populations. Dogs are implicated in the extinction of 11 species and pose threats to over 188 others(45). Panda reserves are often closely linked to human settlements, allowing dogs to roam freely over distances exceeding 10 km overnight; some feral dogs even establish permanent residence within reserves. A comprehensive survey in six village groups surrounding Liziping Reserve found 64% (212/334) of dogs were free-roaming(11). Across all panda reserves, 40% of panda habitat falls within the roaming range of stray dogs. Between 2013 and 2014, dog densities in eight counties surrounding Foping National Nature Reserve (Zhouzhi, Hu, Ningshan, Foping, Yang, Chenggu, Liuba, Taibai) ranged from 3.71 to 20.80 dogs/km^2, densities sufficient to sustain transmission of canine infectious pathogens(26). Since dogs and pandas may share habitats or water sources, the potential for direct or indirect contact between the two species increases, elevating disease transmission risks (46). Canine distemper virus (CDV), a known cause of mortality in pandas, has been transmitted from domestic dogs(26). In Liziping Reserve, 21% of surveyed village dogs tested seropositive for antibodies against at least one of four viruses (CDV, CPV, Rotavirus, Rabies virus) known to be fatal

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to pandas, and many terrestrial mammals within the reserve (e.g., red pandas, yellow-throated martens) are susceptible(18). Beyond viruses, 67% of dog fecal samples were infected with at least one gastrointestinal parasite; hookworms (*Ancylostoma caninum*) and threadworms (*Strongyloides* sp.) found in dogs can also infect pandas. Dogs provide pathways for disease transmission into wildlife populations, potentially causing epidemics or extinctions across multiple species(47).

3.2 Livestock

Field investigations reveal that cattle, sheep, and horses frequently roam freely into protected areas. Livestock grazing has become the second most common anthropogenic disturbance to panda reserves (after logging), affecting 19% of the reserve area in the Minshan Mountains(48). Livestock numbers have increased significantly in some areas over recent years; for instance, livestock numbers in Wanglang National Nature Reserve increased ninefold over 15 years(49). Pathogen transmission between livestock and wildlife at these shared interfaces is often bidirectional. Approximately 77% of livestock pathogens are capable of infecting multiple host species, including wildlife(48, 50). In panda habitats, livestock feces may contaminate water and food sources, substantially raising the risk of pathogen ingestion by pandas, who spend over 14 hours a day feeding. Therefore, the spillover of livestock pathogens to pandas and other wild species poses substantial conservation challenges. Despite these concerns, systematic surveillance of pathogens shared by livestock and wildlife across the network of panda reserves remains insufficient.

3.3 Wild Sympatric Species

Panda nature reserves also serve as sanctuaries for numerous sympatric endangered and vulnerable species. Species benefiting from habitat protection span 6 orders and 18 families, including one bird order (Galliformes) and five mammal orders: Primates (e.g., Sichuan snub-nosed monkey, Proboscis monkey, Tibetan macaque), Carnivora (e.g., Asian black bear, Gray wolf, Red fox, Red panda, Hog badger, Masked palm civet, Asian golden cat, Leopard cat), Artiodactyla (e.g., Forest musk deer, Tufted deer, Sambar deer, Chinese muntjac, Takin, Chinese serow, Chinese goral, Wild boar), Rodentia (e.g., Chinese bamboo rat, Malayan porcupine, Himalayan marmot), and Lagomorpha (e.g., Gansu pika, Plateau pika)(21, 51). This sympatric wildlife shares ecological space and resources with giant pandas, which may facilitate pathogen transmission. However, there has been no systematic investigation or research on the pathogens carried by these animals and their potential impacts. Through non-invasive sampling, high-throughput sequencing analyses targeting viruses, bacteria, and parasites are imperative for establishing a pathogen repository of sympatric species. This is crucial for investigating pathogens within protected areas and characterizing their distribution patterns and pathogenic traits. It is therefore recommended to establish a "sympatric species pathogen inventory program" and enhance monitoring feasibility through non-invasive sampling methods such as fecal or skin shedding collection.

4. Cross-Species Transmission Mechanisms of Pathogens in Giant Panda Reserves

Pathogens can be transmitted among giant pandas and between pandas and other species through various of direct and indirect mechanisms (Figure 1). For example, the roundworm (*B. schroederi*) can infect pandas primarily via fecal–oral transmission: (1) eggs adhere to the feet of pandas after they walk across contaminated ground, and are subsequently ingested when pandas handle bamboo with their forepaws; (2) pandas mark trees using urine or feces, and parasites can be transmitted when another panda rubs its nose or licks the same site. Increased panda density can significantly elevate the risk of such fecal–oral transmission(20). The high diversity of sympatric wildlife species in panda habitats, and their frequent direct or indirect contact, create favorable conditions for interspecies and cross-species pathogen transmission. Intermediate hosts increase the likelihood of cross-species transmission, potentially driving viral adaptation and evolution, enhancing pathogenicity and transmissibility, and posing significant threats(26, 46). For example, several animal groups (camelids, mustelids, canids, procyonids, felids, ursids, viverrids, non-human primates) are highly susceptible to CDV and can act as hosts, facilitating virus transmission between dogs and pandas(52) (27). Stray cats or rodents infected with *T. gondii* could be sources of fatal infections in pandas (53). Additionally, vectors like ticks play a crucial role in disease transmission by exchanging pathogens with vertebrate

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hosts during blood-feeding. Studies show that ticks infesting pandas harbor broad viral genetic diversity and share numerous viruses with pandas(42).

The multi-host nature and host adaptability of pathogens increase the likelihood of cross-species transmission and complicate control efforts. The rich diversity of species within panda reserves and their direct or indirect contacts create favorable conditions for pathogen spread. Carnivore protoparvovirus 1, represented by FPV and CPV, is globally distributed with a broad host range. FPV and CPV infections have been documented in diverse species including raccoons, lions, leopards, giant pandas, and white tigers [25]. *Hepatozoon* spp. are apicomplexan blood parasites capable of infecting various vertebrates globally, including canids, ursids, felids, and many others(16) . Other multi-host pathogens include Rabies virus, Rotavirus, *Hepatozoon* sp., and *Toxoplasma gondii* . Given the close contact among pandas, domestic animals, and sympatric wildlife, it is crucial to investigate the transmission dynamics and genetic diversity of pathogens across hosts. FPV strains detected in fecal samples from stray cats within reserves showed >99% similarity in coding regions to FPV found in giant pandas and red pandas(54). This high genetic relatedness suggests the possibility of cross-species transmission among "giant panda, red panda and cat". Due to rapid evolution and transmission, the host range of some viruses, like parvoviruses and CDV, continues to expand, threatening multiple endangered wildlife species(25, 27). These studies demonstrate potential cross-species transmission events of pathogens between pandas and sympatric animals within the same ecosystem, revealing the genetic diversity and transmission relationships of different viral pathogens within the "panda - sympatric wildlife – arthropod-domestic animal " nexus. Furthermore, beyond known viruses, a metaviromic survey of giant pandas and sympatric animals and vectors (red pandas, stray cats, wild rodents, mosquitoes) within the same habitat identified 250 viral genomes, revealing highly diverse viral communities across multiple animal species in the ecosystem and emphasizing the need to study the prevalence, risk, and significance of carnivore infectious diseases in pandas(54).

5. Conclusions and Future Directions

The effectiveness of nature reserves depends on our understanding and management of the complex interactions among humans, domestic animals, and the wildlife we aim to protect, such as the giant panda. Conservation efforts for giant pandas are now shifting from passive, in situ protection toward more proactive and precise intervention strategies. In the foreseeable future, the conservation status of giant pandas will, to a large extent, depend on the advancement of pathogen surveillance and disease prevention measures. Although considerable progress has been made in understanding animal diseases over the past five decades, our current knowledge of disease processes, especially those occurring at the interface of domestic and wild animals, remains limited. Future efforts should focus on the following aspects:

Quantifying Epidemiological Parameters. The lack **of accurate quantitative data describing** the spatial distribution of**giant panda sympatric animals constrains our understanding of pathogen transmission. Therefore,** future **efforts need to collect essential data for describing disease risks involving humans, domestic animals, and wildlife, including animal species distribution, population sizes and structure, and contact rates between and within different animal species.**

Establishing a Pathogen and Disease Database. Incomplete information on animal pathogens and disease status within reserves, a narrow focus on a limited number of known viruses (e.g., CDV, CPV, influenza A), often targeting specific species rather than the broader host-pathogen network and a growing number of diseases reported**in captive rather than wild pandas limit pathogen research. Therefore, systematically collect samples from livestock (horses, cattle, sheep), domestic pets (dogs, cats), giant pandas, various sympatric wildlife, and vectors (ticks, mosquitoes) at the wildlife-domestic animal interface within panda reserves. Utilize metagenomics and metaviromics sequencing and analysis to establish a wildlife pathogen** community **database for early pathogen discovery and characterization.**

Pathogen Characterization & Monitoring: Analyze pathogen prevalence (occurrence and infection intensity), host range, cross-species transmission potential, pathogenicity, and resistance profiles. Elucidate

the diversity and spatiotemporal distribution of pathogens carried by pandas and sympatric wildlife, along with factors influencing cross-species transmission. Establish robust surveillance systems.

Risk Assessment & Integrated Management: Unravel the bidirectional transmission potential of key pathogens between pandas and domestic animals. Assess spillover/spillback risks. Systematically identify the sources and intensity of disease threats to pandas and domestic animals. Integrate wildlife pathogen surveillance with livestock and human surveillance systems. Develop innovative technologies for enhanced pathogen identification. Utilize identified diseases to guide risk assessment and prioritize investments. Design and implement robust, open-access early warning systems. Improve understanding of current/recent outbreaks. Share information/data and build pathogen surveillance capacity, particularly in wildlife. Address risks from pandemics/epidemics, increased disease risk and susceptibility, and vector-borne disease expansion. Promote early intervention by mitigating the lack of pathogen early warning and surveillance methods. Provide scientific and technological support for wildlife disease prevention and control, species conservation, and biodiversity maintenance.

To effectively manage pathogens within panda reserves, it is crucial to establish targeted early warning systems and surveillance networks at the wildlife–livestock interface for emerging risk identification. Here, we Propose a Workflow for an Early Warning System for Diseases in Giant Pandas and Sympatric Wildlife/Domestic Animals. 1)**Pathogen Baseline Survey.** Collect fecal samples and vectors (ticks, mosquitoes, fleas) from giant pandas, sympatric wildlife (e.g., wild boar, golden snub-nosed monkey, takin, leopard cat), and domestic animals (cattle, sheep, horses, dogs, cats). Employ meta-genomics/-viromics, microscopy, and molecular diagnostics for systematic pathogen screening at the wildlife-domestic animal interface. Utilize bioinformatics to identify viruses, bacteria, and parasites, establishing a reserve pathogen database and characterizing pathogen diversity, including key pathogens and zoonoses. Link pathogen data to individual animal information and spatiotemporal data (field surveys, camera traps, remote sensing, GPS tracking) to analyze distribution patterns and inter-population transmission. 2) **Pathogen Characterization & Transmission.** Analyze key pathogens (e.g., Canine Distemper Virus, antibiotic-resistant *E. coli* , hookworms) for pathogenicity (using VFDB for virulence genes), antimicrobial resistance (AMR; using CARD/ResFinder for resistance genes), cross-species potential, and transmission modes. Assess strain similarity, gene flow/recombination, and host receptor compatibility to identify critical genes for adaptation and predict transmission trends. Map transmission chains and construct host-pathogen interaction networks based on prevalence, vectors, and potential hosts. 3) **Transmission Risk Assessment & Early Warning.** Elucidate bidirectional pathogen transmission between hosts using interaction networks. Develop risk scoring criteria integrating pathogen factors (prevalence, pathogenicity, AMR, transmissibility) and ecological factors (host/vector diversity/density). Build predictive models (e.g., Random Forest, Bayesian) to assess spillover (wildlife-to-domestic/human) and spillback (domestic/human-to-wildlife) risk levels. Systematically identify disease threat sources and intensities to establish a Giant Panda National Park Early Warning System. Identify high-risk spatiotemporal zones for targeted surveillance and control strategies.

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Conflicts of Interest: The authors declare no competing interests.

Table 1: Giant panda viruses

Table 2: Giant panda bacteria

Table 3: Giant panda parasites

Table 4: Cross - species susceptible animals (non - giant pandas) to common pathogens of giant pandas

Figure 1. Pathways of multi-host pathogen transmission among giant pandas, sympatric wildlife, and domestic animals in nature reserves.

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backend=biber, style=numeric, sorting=none]biblatex **References**

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Table 1: **Giant panda viruses**

Pathogens	Classification	Infection site
Ailuropoda melanoleuca Virus (Aimelvirus)	N/A	Intestine
Giant Panda Gemycircularvirus 2 (GpGMCV2)	*Gemycircularvirus (#)*	Nasopharynx, intestine, blood
Panda-stool-associated RNA Virus (PANSAVIRUS)	N/A	Intestine
Giant panda gemycircularvirus (GPGMCV)	*Gemycircularvirus (#)*	Intestine, Nasopharynx
Giant panda anellovirus (GPAV)	Anelloviridae *(*)*	Blood/Multiple tissues
Giant Panda Circovirus-like Virus 1 (GPCV1)	Circoviridae (*)	Blood/Multi tissue
Ailuropoda melanoleuca Papillomavirus 1 (AMPV1)	Papillomaviridae *(*)*	Nasopharyngeal
Picobirnavirus-like	*Picobirnavirus (#)*	Intestine
Giant panda polyomavirus 1 (GPPYV1)	Polyomaviridae (*)	Nasal cavity/Intestine

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Giant panda-associated circovirus	*Circovirus (#)*	Blood (speculated)
Influenza A (H1N1) pdm09 Virus	*Influenza (#)*	Respiratory tract
Rabies virus	*Lyssavirus (#)*	Nervous system
Influenza A Virus	*Alphainfluenzavirus (#)*	Respiratory tract
Rotavirus	*Rotavirus (#)*	Intestine
Feline panleukopenia virus (FPV)	*Parvovirus (#)*	Systemic (mainly affecting the intestines a
Canine parainfluenza virus (CPIV)	*Respirovirus (#)*	Respiratory tract
Canine coronavirus (CCV)	*Alphacoronavirus (#)*	Intestine
Canine distemper virus (CDV)	*Morbillivirus (#)*	Heart, liver, spleen, lungs, kidneys, intesti
Canine parvovirus (CPV)	*Parvovirus (#)*	Intestine
Canine adenovirus (CAV)	*Mastadenovirus (#)*	Liver
Pseudorabies virus(PRV)	Herpesvirus (#)	N/A

N/A: not available; (*): be classified into Family; (#): Indicates the genus; VNT: Virus neutralization assay; PCR: Polymerase chain reaction; n-PCR: Nested-polymerase chain reaction; $TCID_{50}$: Tissue culture infective dose; HA: hemagglutination assay; HI: Hemagglutination-inhibition; IFA: immunofluorescence assay; qpcr: quantitative PCR; SN: Serum neutralization; ELISA: Enzyme-linked immunosorbent assay, EM: Electron microscopy, IEM: Immuno-electronmicroscopy, FAT: Fluorescent antibody tests; NGS: Next-Generation Sequencing.

Table 2: **Giant panda bacteria**

Pathogens	Classification	Infection site
Acinetobacter nosocomialis	*Acinetobacter (#)*	Intestine
Citrobacter portucalensis	*Belonging to a species of Citrobacter (#)*	Intestine
Proteobacteria	Proteobacteria(+)	Oral cavity
Oxalobacteraceae	Proteobacteria(+)	Oral cavity
Actinobacillus	*Actinobacillus (#)*	Oral cavity
Firmicutes	Firmicutes(+)	Oral cavity
Moraxella	*Moraxella (#)*	Oral cavity
Bacteroidota	*Bacteroidetes (#)*	Oral cavity
Clostridium	*Clostridium (#)*	Intestine
Propionibacterium	*Propionibacterium (#)*	Skin
Pseudomonas	*Pseudomonas (#)*	Skin
Arthrobacter	*Arthrobacter (#)*	Skin
Streptococcus	*Streptococcus (#)*	Skin
EIEC O152	*Escherichia (#)*	Intestine
Staphylococcus aureus	*Staphylococcus (#)*	Lungs
Salmonella enterica	*Salmonella (#)*	Intestine
Helicobacter	*Helicobacter (#)*	Intestine
Shigella	*Shigella (#)*	Intestine
Hafnia alvei	*Hafnia (#)*	Intestine
Flavobacterium	*Flavobacterium (#)*	Intestine
Lactococcus	*Lactococcus (#)*	Intestine

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Yersinia enterocolitica	*Yersinia (#)*	Intestine
Klebsiella pneumoniae	*Klebsiella (#)*	Intestine, blood, urinary and reproductive tract
Campylobacter jejuni	*Campylobacter (#)*	Intestine
Proteus mirabilis	*Proteus (#)*	Urethra and wounds
Fusarium	*Fusarium (#)*	Intestine
Aspergillus	*Aspergillus (#)*	Intestine
Enterococcus	*Enterococcus (#)*	Intestine
Cladosporium cladosporioides	*Cladosporium (#)*	Skin (nose and back)

N/A: not available; (+):be classified into phylum; (-):be classified into order; (#): Indicates the genus; mags: metagenome-assembled genomes; PCR: Polymerase chain reaction;

Table 3: **Giant panda parasites**

Pathogens	Classification	Infection site	Signs
Eimeria sp.	*Eimeria (#)*	Intestine	N/A
Sarcocystis sp.	*Sarcocystis (#)*	Muscle	No specific s
Stilesia sp.	*Stilesia (#)*	Intestine	N/A
Ogmocotyle indica	*Ogmocotyle (#)*	*Intestine*	Intestinal w
Haemaphysalis aponommoides	*Haemaphysalis (#)*	Body surface	Anemia, ski
Demodex ailuropodae	*Demodex (#)*	Hair follicles, sebaceous glands	Itching, nod
Haemaphysalis ailuropodae	*Haemaphysalis (#)*	Body surface	Anemia, ski
Strongyloides sp.	*Strongyloides (#)*	Small intestine	N/A
Ixodes ovatux	*Ixodes (#)*	Body surface	Swelling, itc
Ixodes acutitarsus	*Ixodes (#)*	Body surface	Swelling, itc
Dermacentor taiwanensis	*Dermacentor (#)*	Body surface	N/A
Haemaphysalis warburtoni	*Haemaphysalis (#)*	Body surface	Swelling, itc
Toxascaris seleactis	*Toxascaris (#)*	Intestine	Abdominal
Chaetopsylla ailuropodae	*Chaetopsylla (#)*	Body surface	Anemia, ski
Haemaphysalis apomarmoides	*Haemaphysalis (#)*	Body surface	Swelling, itc
Cryptosporidium sp.	*Cryptosporidium (#)*	Intestine	Waterlike di
Cryptosporidium andersoni	*Cryptosporidium (#)*	Intestine	Diarrhea an
Babesia sp.	*Babesia (#)*	Erythrocyte	Fever, anem
Baylisascaris schroederi	*Baylisascaris (#)*	Pancreatic duct, digestive tract	Weight loss,
Enterocytozoon bieneusi	*Enterocytozoon (#)*	Small intestines	Chronic dia
Ancylostoma ailuropodae	*Ancylostoma (#)*	Intestine	Bloody stoo
Chorioptes panda	*Chorioptes (#)*	Skin surface	Skin inflam
Cryptosporidium giant panda genotype	*Cryptosporidium (#)*	Intestine	Diarrhea an
Hepatozoon sp.	*Hepatozoon (#)*	Blood	N/A
Toxoplasma gondii	*Toxoplasma (#)*	Intestine and lungs	Acute gastr
Clonorchis sinensis	*Clonorchis (#)*	Liver	N/A
Trichuris trichiura	*Trichuris (#)*	Intestine	N/A
Demodex sp.	*Demodex (#)*	Skin	Hair remova
Trichinella	*Trichinella (#)*	Muscle	N/A
Blastocystis sp.	*Blastocystis (#)*	Intestine	N/A
Ancylostoma caninum	*Ancylostoma (#)*	Small intestine	Inflammatio
Lungworm	*Nematoda or Platyhelminthes (+)*	Intestines, lungs	
Haemaphysalis hystricis	*Haemaphysalis (#)*	Body surface	
Haemaphysalis flava	*Haemaphysalis (#)*		
Haemaphysalis megaspinosa	*Haemaphysalis (#)*		

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Ixodes granulatus	*Ixodes (#)*		
Chaetopsylla mikado	*Chaetopsylla (#)*		
Haemaphysalis longicornis	*Haemaphysalis (#)*		
Helictometra sp.	*Helictometra (#)*	Intestine	N/A
Ancylostoma caninum	*Ancylostoma (#)*	Small intestine	Causes blee
Toxocara canis	*Toxocara (#)*	Intestine	N/A
Tyzzeria sp.	*Protozoa (#)*	N/A	N/A
Bunostomum sp.	*Nematode (#)*	Intestine	N/A
			cause anemi

N/A: not available; (+):be classified into phylum; (#): Indicates the genus; SS: saline smear; SI: iodine staining; PCR: Polymerase chain reaction; NGS: Next-Generation Sequencing.

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Table 4: Cross - species susceptible animals (non - giant pandas) to common pathogens of giant pandas

Patpigens	Hosts
Polyomavirus	Domestic animals: cattle(1), alpaca(2) Pet: dog, cat(3) Economic animals: rabbit(1) V
Canine coronavirus (CCV)	Pet: dog(7), cat(8) Wild animals: vulpes, raccoon dog(9)
Rabies virus	Domestic animals: cattle(10), sheep and goat(11), horse, donkey(12) Pet: dog(11), cat
Picobirnavirus-like	Domestic animals: pig, cattle, horse(15) Pet: dog(15) Poultry: chicken(16) Economic a
Influenza A virus	Domestic animals: pig, horse(18) Poultry: duck(18), chicken, turkey, goose(19) Wild a
Rotavirus	Domestic animals: cattle(21), pig, sheep and goat, horse(22) Pet: dog, cat(22) Poultry
Feline panleukopenia virus (FPV)	Pet: cat(25) Economic animals: breeding mink(26) Wild animals: panthera uncia(25),
Canine parainfluenza virus (CPIV)	Domestic animals: pig(29), cattle(30) Pet: dog, cat(30) Wild animals: red panda(29),
Canine distemper virus (CDV)	Domestic animals: pig(32) Pet: dog(33), cat(32) Wild animals: raccoon, red fox(34), r
Canine parvovirus (CPV)	Pet: dog(37) Wild animals: gray wolf, suburban wolf, jackal, gray fox, vulpes macrotis
Canine adenovirus (CAV)	Domestic animals: pig, goat, sheep(41) Pet: dog(42) Wild animals: rodents, primates(
Papillomavirus	Domestic animals: cattle(45) Pet: dog(46), cat(47) Economic animals: rabbit(48) Wil
Pseudorabies virus (PRV)	Domestic animals: pig(51), goat, cattle(52) Pet: dog(51), cat(52) Wild animals: jagua
Circovirus	Domestic animals: pig, cattle, goat(58) Pet: dog, cat(58) Economic animals: rabbit(58

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Eimeria sp	Domestic animals: cattle(64), sheep and goat(65), alpaca, donkey, camel(66), pig(67)
Babesia sp	Domestic animals: cattle, goat(72), sheep(73) Pet: dog(74) Wild animals: Skunk, racc
Haemaphysalis kitaokai	Domestic animals: cattle(75), horse(76) Wild animals: sika deer, pig, raccoon(77), rod
Enterocytozoon bieneusi	Domestic animals: pig(80), cattle, sheep, horse(81) Pet: dog, cat(82) Poultry: chicken
Haemaphysalis spinigera	Domestic animals: cattle(88), sheep(89) Pet: dog(90) Wild animals: asian house shrew
Lungworm	Domestic animals: sheep and goat(94), cattle(95), camel, horse(96) Pet: cat(97), dog(9
Hepatozoon sp	Pet: dog(99), cat(100) Poultry: guinea fowl(101) Economic animals: rabbit(102) Wild
Toxoplasma gondii	Domestic animals: house(106), sheep, goat, pig, cattle, yak(107), camel(108) Pet: dog(
Haemaphysalis hystricis	Domestic animals: pig, water buffalo(112) Pet: dog(112) Wild animals: badger(113), t
Haemaphysalis flava	Domestic animals: pig, horse, sheep and goat, cattle(115) Pet: dog(116), cat(117) Wil
Clonorchis sinensis	Domestic animals: pig, cattle(118) Pet: dog(119), cat(120) Poultry: chicken, duck(118
Haemaphysalis megaspinosa	Domestic animals: house(121) Wild animals: sika deer(122), pig, raccoon(77)
Strongyloides sp.	Domestic animals: pig, horse(123), cattle, sheep and goat(124) Pet: dog(123), cat(125
Ixodes granulatus	Pet: cat(128) Wild animals: anderson's house mouse, black rat(129), rodents(128), tre
Ixodes ovatus	Domestic animals: house(133) Pet: dog(133), cat(134) Wild animals: hare(133), forest
Trichuris sp	Domestic animals: camel, sheep(136), pig(137) Pet: dog(136) Wild animals: deer(136)
Helictometra sp.	Domestic animals: sheep, goat(139), cattle(140)
Toxocara canis	Domestic animals: pig(141) Pet: dog(142) Poultry: chicken(143) Economic animals: ra
Ancylostoma caninum	Pet: dog(145), cat(146) Wild animals: wild dog(147), rodents(148), rhesus monkey(14
Demodex sp.	Domestic animals: cattle, sheep and goat(149), pig, horse(150) Pet: dog(151), cat(149
Ixodes acutitarsus	Wild animals: komodo dragon, tree shrew, pig(154), taiwan black bear(155)
Stilesia sp.	Domestic animals: sheep(156), camel(157), goat(157) Wild animals: water antelope, b
Dermacentor taiwanensis	Domestic animals: cattle(159) Pet: dog(160) Wild animals: pig, black bear(155), rode
Tyzzeria sp.	Poultry: duck(162) Economic animals: rabbit(163) Wild animals: grey goose(164), sw
Trichinella	Domestic animals: pig(168), horse(169), sheep(170) Pet: dog(170), cat(170) Poultry: c
Blastocystis sp.	Domestic animals: pig, cattle, sheep and goat(174) Pet: dog, cat(174) Poultry: chicke
Bunostomum sp.	Domestic animals: sheep(177), cattle(178) Wild animals: cervus elaphus sibiricusr(179
Cryptosporidium sp.	Domestic animals: cattle, sheep and goat, pig, horse(180), donkey(181), yak, goat(182
Chaetopsylla mikado	Pet: cat(185) Wild animals: otter, mink(185)
Haemaphysalis longicornis	Domestic animals: sheep, goat, horse, cattle(186), pig(187) Pet: dog(188), cat(189) Po
Haemaphysalis aponommoides	Domestic animals: goat(191), sheep, cattle(192), horse(193) Pet: dog(193) Wild anima
Acinetobacter nosocomialis	Pet: dog, cat(194) Wild animals: birds, fishes(195)
Proteobacteria	Domestic animals: cattle(196), sheep(197), pig(198) Pet: dog(198) Poultry: chicken(19
Propionibacterium	Domestic animals: cattle(199), sheep and goat(200), pig(201) Pet: dog(200) Poultry: c
Burkholderia	Domestic animals: house(203), goat, sheep(204)donkey, mule(205), camel(206) Pet: do
Oxalobacter	Domestic animals: pig, cattle, sheep and goat(208), horse(209) Pet: cat(210) Economi
Citrobacter portucalensis	Pet: dog, cat(212) Poultry: chicken(213) Wild animals:,axolotl(214), spottedturtle(215
Enterococcus	Domestic animals: pig, cattle, donkey(216) Pet: dog(217), cat(216) Poultry: pigeon, ch
Salmonella enterica	Domestic animals: pig, cattle(221) Poultry: chicken(222), turkey(223) Wild animals: h
Actinobacillus	Domestic animals: pig, sheep and goat(226), cattle, horse(227) Pet: cat(226) Wild ani
Klebsiella pneumoniae	Domestic animals: pig(230), cattle(231) Pet: dog, cat(232) Poultry: chicken(230) Ecor
Hafnia alvei	Domestic animals: house, sheep, goat, cattle(233), camel(234) Pet: dog(235), cat(234)
Firmicutes	Domestic animals: cattle(237), pig(238), donkey, goat, sheep, horse(239) Pet: dog(238
Flavobacterium	Poultry: chicken(246) Economic animals: freshwater aquaculture fish(247), salmon(248
Pseudomonas	Domestic animals: cattle, horse(250), sheep and goat(251) Pet: dog(250) Poultry: goo
Arthrobacter	Domestic animals: cattle, pig(254) Economic animals: rabbit(255)
Staphylococcus aureus	Domestic animals: cattle(256), sheep and goat(257), horse, pig(258) Pet: dog(256) Pou
Campylobacter jejuni	Domestic animals: cattle, sheep and goat(261), pig(262) Pet: dog, cat(263) Poultry: b
Streptococcus	Domestic animals: cattle, pig, horse(264), sheep(265) Pet: dog(264), cat(266) Poultry:
Helicobacter	Domestic animals: pig(268), horse(269), cattle, goat(269) Pet: dog, cat(270) Poultry:
Moraxella	Domestic animals: cattle(275), horse(276), goat, sheep(277) Pet: dog, cat(278) Wild a

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Bacteroidetes	Domestic animals: sheep and goat, pig(280), cattle(281) Pet: dog(282) Poultry: chicke
Proteus mirabilis	Domestic animals: cattle(285) Pet: dog(286), cat(287) Poultry: chicken(288) Economi
Lactococcus	Domestic animals: cattle(291), horse, pig(292), camel(293) Pet: dog, cat(292) Poultry:
Clostridium	Domestic animals: cattle(297), sheep and goat, pig(298), horse(299) Pet: dog, cat(298
Yersinia enterocolitica	Domestic animals: pig, cattle(301), sheep and goat, horse(302) Pet: dog, cat(302) Wil
Shigella	Domestic animals: cattle(304) Pet: dog(305) Poultry: chicken(306) Economic animals:
Fusarium	Domestic animals: house(309), pig(310) Pet: dog(309) Poultry: quail, chicken(311) Ec
Aspergillus	Domestic animals: dairy cattle(313), pig, horse(314) Pet: dog(315) Poultry: turkey(31
Cladosporium cladosporioides	Domestic animals: sheep(316) Pet: dog(317), cat(318) Wild animals: panthera uncia(3

N/A: not available; HT: Horizontal transmission; VT: Vertical transmission; cattle: GD: Global distribution;

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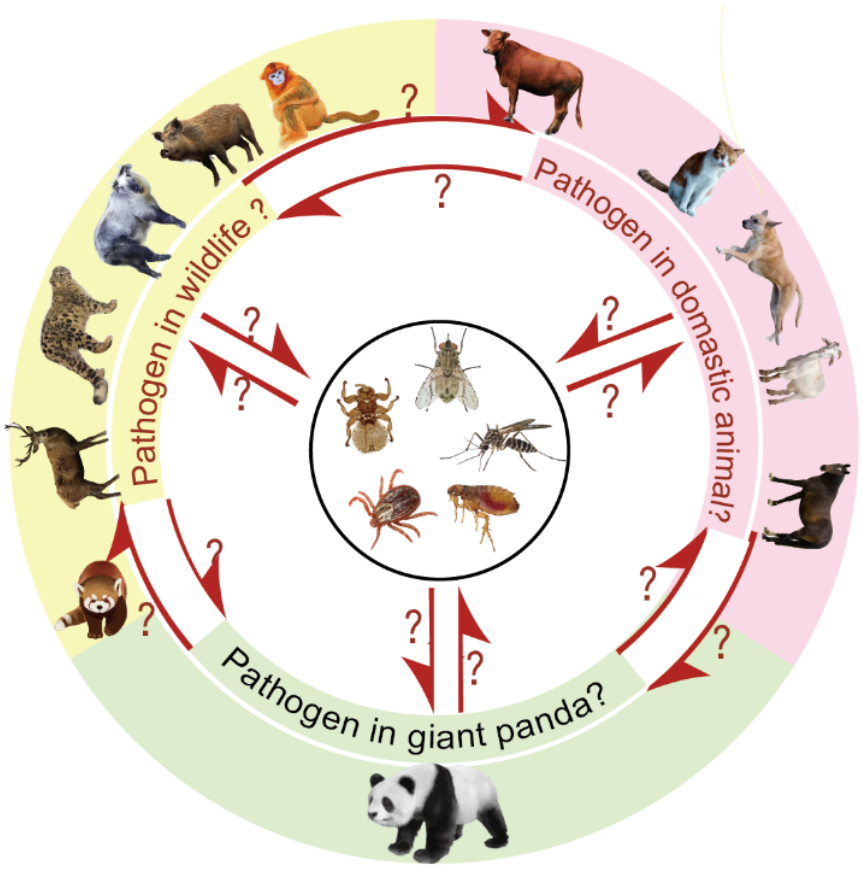


Figure 1. Pathways of multi-host pathogen transmission among giant pandas, sympatric wildlife, and domestic animals in nature reserves.